Comprehensive Data Analysis of Block Test Results

Experimental mixes Exp 1 – Exp 4 versus conventional Concrete Hollow Block (CHB) · Source: Stat_Coaching.xlsx · 5 groups × 7 replicates × 8 tests (280 observations) · $\alpha = 0.05$

Executive summary

1. Data quality is good. No missing values; Grubbs' test finds no outliers in any of the 40 test × group samples; residuals are close to normal; variances are equal among Exp 1–4 for every test. Results from parametric, robust (Welch, Games–Howell) and non-parametric (Kruskal–Wallis) methods agree.

2. All Exp mixes outperform CHB on every property (Dunnett $p < 0.0001$ for all 32 comparisons): ~67% lighter, 3.2–4.0× the 28-day compressive strength, 2.5–2.9× the screw pull-out, 3.1–3.6× the split tensile strength and 35–38% lower moisture absorption.

3. Exp 4 is the weakest Exp mix. It is significantly lower in compressive strength at 28 days than Exp 1, 2 and 3 (28-day $\eta^2 = 0.58$), lowest in screw pull-out, and it ranks last in 96% of random weightings of the decision criteria.

4. Exp 1, Exp 2 and Exp 3 are statistically equivalent in 28-day strength. They differ in secondary traits: Exp 2 has the best balance, Exp 1 the highest strength and fastest strength gain, and Exp 3 the lowest water absorption and the most uniform results.

5. Recommended mix: Exp 2. It has the highest composite score (0.85) and ranks first in 74% of 20,000 random weightings; Exp 3 is the runner-up (23%).

6. Benchmarks: Exp mixes reach 10.1–12.5 MPa at 28 days, above ASTM C129 (non-load-bearing) and the commonly cited PNS 16 load-bearing level, but below the ASTM C90 load-bearing average of 13.8 MPa. CHB (3.15 MPa) falls below ASTM C129.

7. Sample size: $n = 7$ gives ≥ 84% power for compressive strength, screw pull-out and moisture absorption, but only 47% power for split tensile strength; about 14 specimens per mix would be needed to confirm a difference in split tensile strength.

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1. Data Overview and Methodology

Item	Description
Groups	Exp 1, Exp 2, Exp 3, Exp 4 (experimental block mixes) and CHB (conventional concrete hollow block, control)
Replicates	7 specimens per group per test (balanced design); no missing values
Physical tests	Weight in air (kg), unit weight (kg/m³), moisture absorption (%)
Mechanical tests	Compressive strength at 7, 14 and 28 days (psi), screw pull-out (kg), split tensile strength (MPa)
Design	Completely randomised, one factor (block type, 5 levels); curing age as a second factor for compressive strength
Software / method	Python (SciPy, statsmodels, scikit-learn); all-group results reproduce the workbook ANOVA and Tukey HSD/Kramer sheets
Significance level	$\alpha = 0.05$ (two-sided); family-wise error controlled by Tukey, Games–Howell and Dunnett procedures

Analysis workflow

Step	Question answered	Methods
Data quality	Is the data clean and are test assumptions met?	Grubbs' test, IQR rule, Shapiro–Wilk, Levene, Bartlett, Q–Q plots
Description	What are the typical values and how variable are they?	Mean, SD, CV, median, IQR, skewness, kurtosis, t- and bootstrap CIs
Group differences	Do block types differ, and which ones?	One-way ANOVA, Welch ANOVA, Kruskal–Wallis, Tukey HSD, Games–Howell, Dunnett
Magnitude & power	How large are differences and was n large enough?	η^2, ω^2, Cohen's d, Cohen's f, power, minimum detectable difference
Strength over time	How does strength develop with curing age?	Two-way ANOVA, logarithmic and ACI 209-type hyperbolic models
Relationships	Which properties move together; can one predict another?	Pearson and Spearman correlation, simple and multiple regression
Multivariate view	Do mixes differ when all properties are considered together?	MANOVA, PCA, linear discriminant analysis, hierarchical clustering
Decision	Which mix is best overall and how robust is the choice?	Benchmarking, efficiency ratios, composite index, scenario and Monte Carlo sensitivity

Raw data

Test / Group	Replicate values (1 → 7)
Weight in Air (kg)	Exp 1: 4.650, 4.620, 4.660, 4.670, 4.660, 4.690, 4.650
	Exp 2: 4.610, 4.640, 4.670, 4.630, 4.630, 4.680, 4.600
	Exp 3: 4.690, 4.600, 4.630, 4.670, 4.620, 4.660, 4.670
	Exp 4: 4.670, 4.630, 4.690, 4.680, 4.620, 4.650, 4.610
	CHB: 14.100, 13.900, 14.400, 14.200, 14.000, 14.300, 14.200
Unit Weight (kg/m³)	Exp 1: 877.1, 871.5, 879.0, 880.9, 879.0, 884.7, 877.1
	Exp 2: 869.6, 875.2, 880.9, 873.4, 873.4, 882.8, 867.7
	Exp 3: 884.7, 867.7, 873.4, 880.9, 871.5, 879.0, 880.9
	Exp 4: 880.9, 873.4, 884.7, 882.8, 871.5, 877.1, 869.6
	CHB: 1235.0, 1232.0, 1240.0, 1237.0, 1233.0, 1238.0, 1237.0
Compressive 7 d (psi)	Exp 1: 1122, 992, 1084, 1211, 981, 976, 1166
	Exp 2: 1232, 1350, 1007, 1154, 1096, 1222, 1195
	Exp 3: 1214, 1008, 1158, 1092, 1256, 1124, 1136
	Exp 4: 1045, 946, 992, 954, 1125, 985, 968
	CHB: 192, 183, 201, 210, 187, 189, 195
Compressive 14 d (psi)	Exp 1: 1643, 1348, 1591, 1722, 1362, 1355, 1584
	Exp 2: 1708, 1688, 1455, 1385, 1352, 1421, 1395
	Exp 3: 1504, 1225, 1436, 1285, 1432, 1354, 1348
	Exp 4: 1235, 1102, 1284, 1087, 1452, 1188, 1122
	CHB: 381, 395, 370, 385, 387, 377, 365
Compressive 28 d (psi)	Exp 1: 1972, 1624, 1827, 2030, 1668, 1653, 1871
	Exp 2: 1929, 1943, 1668, 1711, 1697, 1832, 1786
	Exp 3: 1842, 1595, 1769, 1581, 1711, 1668, 1697
	Exp 4: 1581, 1363, 1523, 1349, 1668, 1378, 1392
	CHB: 485, 492, 420, 434, 472, 445, 452
Screw Pull-out (kg)	Exp 1: 28.0, 24.0, 27.0, 29.0, 24.0, 24.0, 27.0
	Exp 2: 32.0, 31.0, 25.0, 26.0, 25.0, 28.0, 27.0
	Exp 3: 29.0, 24.0, 26.0, 23.0, 28.0, 27.0, 27.0
	Exp 4: 24.0, 22.0, 25.0, 23.0, 26.0, 24.0, 22.0
	CHB: 11.0, 9.0, 10.0, 12.0, 7.0, 8.0, 9.0
Split Tensile (MPa)	Exp 1: 0.450, 0.320, 0.430, 0.470, 0.330, 0.330, 0.420
	Exp 2: 0.460, 0.440, 0.330, 0.340, 0.330, 0.360, 0.350
	Exp 3: 0.430, 0.370, 0.420, 0.380, 0.410, 0.360, 0.360
	Exp 4: 0.370, 0.320, 0.360, 0.310, 0.380, 0.330, 0.320

Test / Group	Replicate values (1 → 7)
	CHB: 0.110, 0.090, 0.130, 0.100, 0.080, 0.140, 0.120
Moisture Absorption (%)	Exp 1: 8.50, 9.10, 8.70, 8.20, 8.80, 8.90, 8.40
	Exp 2: 8.10, 8.20, 8.90, 8.30, 8.10, 8.50, 8.40
	Exp 3: 8.30, 8.10, 8.00, 8.40, 8.40, 8.10, 8.20
	Exp 4: 8.60, 8.50, 8.70, 8.80, 8.40, 8.30, 8.90
	CHB: 13.10, 12.70, 14.20, 13.50, 12.90, 13.70, 13.20

2. Data Quality and Assumption Checks

A. Outlier screening (Grubbs' test, two-sided, α = 0.05; critical G = 2.02 for n = 7)

Test	Exp 1: G	Exp 2: G	Exp 3: G	Exp 4: G	CHB: G	IQR-rule flags
Weight in Air (kg)	1.74	1.46	1.50	1.29	1.50	Exp 1: 4.620, 4.690
Unit Weight (kg/m³)	1.74	1.46	1.50	1.29	1.41	Exp 1: 871.5, 884.7
Compressive 7 d (psi)	1.42	1.58	1.65	1.94	1.76	Exp 4: 1125
Compressive 14 d (psi)	1.32	1.50	1.50	1.88	1.46	none
Compressive 28 d (psi)	1.38	1.33	1.60	1.62	1.39	none
Screw Pull-out (kg)	1.35	1.52	1.54	1.53	1.50	none
Split Tensile (MPa)	1.21	1.61	1.36	1.38	1.39	none
Moisture Absorption (%)	1.47	1.93	1.36	1.39	1.70	none

G = largest standardised deviation in the group. All 40 values are below 2.02, so **no statistical outliers** are present. The IQR rule marks four mild values, none confirmed by Grubbs; all data were retained.

B. Normality within each group (Shapiro–Wilk p-values)

Test	Exp 1	Exp 2	Exp 3	Exp 4	CHB	Residuals (all)	Residuals (Exp)
Weight in Air (kg)	0.744	0.617	0.641	0.602	0.958	0.000	0.288
Unit Weight (kg/m³)	0.744	0.618	0.641	0.602	0.804	0.519	0.289
Compressive 7 d (psi)	0.293	0.969	0.968	0.116	0.740	0.785	0.994
Compressive 14 d (psi)	0.117	0.031	0.909	0.296	0.976	0.098	0.050
Compressive 28 d (psi)	0.368	0.352	0.819	0.147	0.785	0.375	0.128
Screw Pull-out (kg)	0.100	0.229	0.686	0.591	0.958	0.343	0.328
Split Tensile (MPa)	0.109	0.027	0.204	0.252	0.949	0.401	0.262
Moisture Absorption (%)	0.986	0.226	0.420	0.949	0.872	0.611	0.865

Shaded cells: p < 0.05 (departure from normality). Only 2 of 40 group samples are flagged, both marginally; the pooled residuals of the Exp-only models are normal for every test. The all-group weight-in-air residuals are non-normal because the CHB scale (14 kg) is three times larger than the Exp scale.

C. Homogeneity of variance (p-values)

Test	Levene (all)	Bartlett (all)	Levene (Exp)	Bartlett (Exp)	Verdict
Weight in Air (kg)	0.0012	< 0.0001	0.5846	0.7863	Equal among Exp; CHB differs
Unit Weight (kg/m³)	0.3587	0.4079	0.5853	0.7867	Equal
Compressive 7 d (psi)	0.0558	0.0003	0.5443	0.6277	Equal among Exp; CHB differs
Compressive 14 d (psi)	0.1187	< 0.0001	0.7616	0.6932	Equal among Exp; CHB differs
Compressive 28 d (psi)	0.0642	0.0097	0.4773	0.5906	Equal among Exp; CHB differs
Screw Pull-out (kg)	0.6856	0.6230	0.6308	0.5446	Equal
Split Tensile (MPa)	0.1890	0.0507	0.3016	0.1324	Equal
Moisture Absorption (%)	0.1608	0.0676	0.5230	0.4227	Equal

Where CHB variance differs from the Exp variance (shaded), the Welch ANOVA and Games–Howell results in Sections 4–5 are the reference; they lead to the same conclusions.

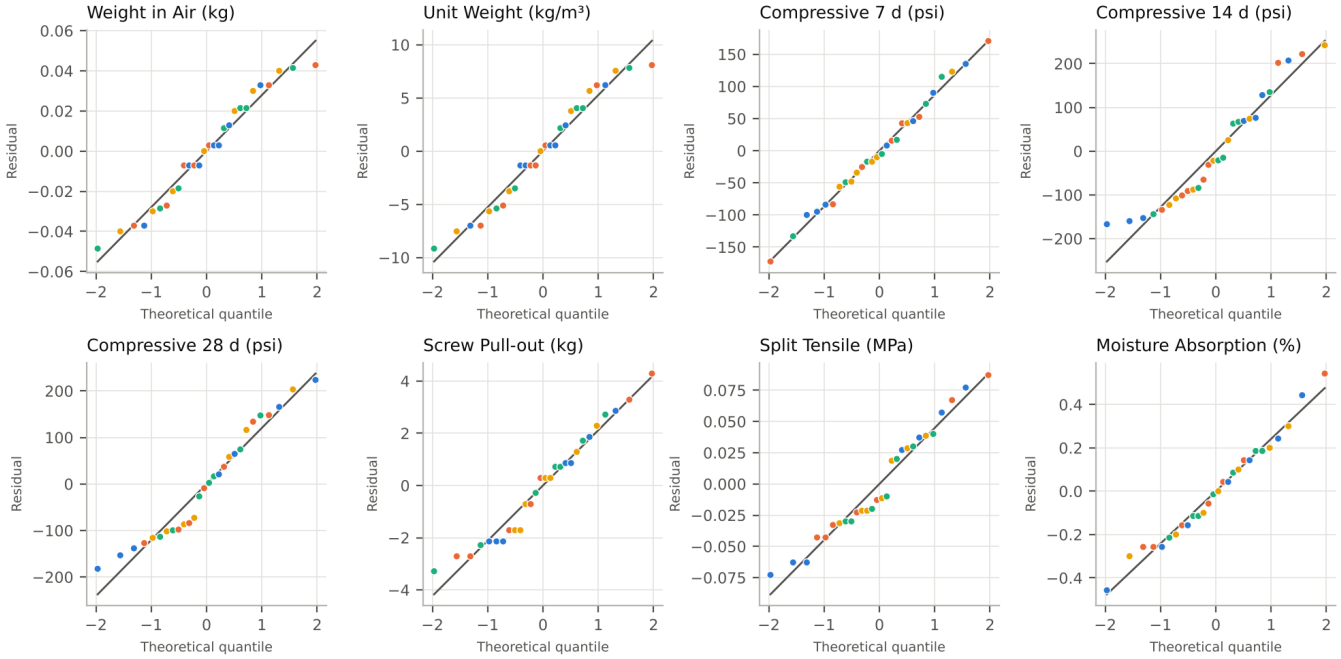


Figure 1. Normal Q–Q plots of residuals (Exp 1–4). Points close to the line indicate normally distributed residuals. Colours identify the mix.

3. Descriptive Statistics

Test	Group	Mean	SD	CV %	Median	IQR	Skew	Kurt.	95% CI (t)	95% CI (bootstrap)
Weight in Air (kg)	Exp 1	4.657	0.021	0.5	4.660	0.015	-0.34	1.53	4.637–4.677	4.641–4.671
	Exp 2	4.637	0.029	0.6	4.630	0.035	0.39	-0.96	4.610–4.664	4.617–4.657
	Exp 3	4.649	0.032	0.7	4.660	0.045	-0.36	-1.26	4.619–4.679	4.626–4.670
	Exp 4	4.650	0.031	0.7	4.650	0.050	-0.00	-1.87	4.621–4.679	4.629–4.671
	CHB	14.157	0.172	1.2	14.200	0.200	-0.17	-0.64	13.998–14.316	14.043–14.271
Unit Weight (kg/m³)	Exp 1	878.5	4.0	0.5	879.0	2.8	-0.34	1.53	874.7–882.2	875.5–881.2
	Exp 2	874.7	5.5	0.6	873.4	6.6	0.39	-0.96	869.6–879.8	870.9–878.5
	Exp 3	876.9	6.1	0.7	879.0	8.5	-0.36	-1.26	871.2–882.5	872.5–880.9
	Exp 4	877.1	5.9	0.7	877.1	9.4	0.00	-1.87	871.7–882.5	873.1–881.2
	CHB	1236.0	2.8	0.2	1237.0	3.5	-0.19	-0.94	1233.4–1238.6	1234.0–1237.9
Compressive 7 d (psi)	Exp 1	1076	95.4	8.9	1084	158	0.23	-1.83	988–1164	1014–1142
	Exp 2	1179	109.0	9.2	1195	102	-0.10	0.52	1079–1280	1106–1255
	Exp 3	1141	80.9	7.1	1136	78.0	-0.25	0.30	1066–1216	1084–1195
	Exp 4	1002	63.2	6.3	985	57.5	1.48	1.85	944–1061	965–1049
	CHB	194	9.2	4.7	192	10.0	0.86	0.35	185–202	188–201
Compressive 14 d (psi)	Exp 1	1515	156.3	10.3	1584	258	-0.04	-2.17	1370–1660	1404–1627
	Exp 2	1486	148.2	10.0	1421	182	1.05	-0.95	1349–1623	1393–1592
	Exp 3	1369	95.9	7.0	1354	118	-0.16	-0.71	1280–1458	1305–1432
	Exp 4	1210	128.9	10.7	1188	148	1.17	1.17	1091–1329	1132–1305
	CHB	380	10.3	2.7	381	12.5	-0.12	-0.62	370–390	373–387
Compressive 28 d (psi)	Exp 1	1806	162.3	9.0	1827	261	0.20	-1.83	1656–1956	1699–1916
	Exp 2	1795	111.1	6.2	1786	176	0.35	-1.71	1692–1898	1722–1869
	Exp 3	1695	92.3	5.4	1697	108	0.34	-0.45	1609–1780	1633–1757
	Exp 4	1465	125.7	8.6	1392	182	0.76	-1.16	1349–1581	1382–1556
	CHB	457	26.8	5.9	452	39.0	0.02	-1.46	432–482	439–475
Screw Pull-out (kg)	Exp 1	26.1	2.1	8.1	27.0	3.5	0.04	-2.07	24.2–28.1	24.7–27.6
	Exp 2	27.7	2.8	10.1	27.0	4.0	0.72	-1.13	25.1–30.3	25.9–29.9
	Exp 3	26.3	2.1	8.1	27.0	2.5	-0.52	-0.77	24.3–28.3	24.9–27.7
	Exp 4	23.7	1.5	6.3	24.0	2.0	0.26	-0.97	22.3–25.1	22.7–24.7
	CHB	9.4	1.7	18.2	9.0	2.0	0.17	-0.64	7.8–11.0	8.3–10.6
Split Tensile (MPa)	Exp 1	0.393	0.064	16.3	0.420	0.110	-0.15	-2.39	0.334–0.452	0.347–0.437
	Exp 2	0.373	0.054	14.5	0.350	0.065	1.12	-0.69	0.323–0.423	0.339–0.414
	Exp 3	0.390	0.029	7.5	0.380	0.050	0.33	-2.09	0.363–0.417	0.370–0.411
	Exp 4	0.341	0.028	8.2	0.330	0.045	0.37	-2.02	0.316–0.367	0.323–0.363
	CHB	0.110	0.022	19.6	0.110	0.030	0.00	-1.20	0.090–0.130	0.096–0.126
Moisture Absorption (%)	Exp 1	8.66	0.310	3.6	8.70	0.40	-0.10	-0.81	8.37–8.94	8.44–8.86
	Exp 2	8.36	0.282	3.4	8.30	0.30	1.30	1.73	8.10–8.62	8.19–8.57
	Exp 3	8.21	0.157	1.9	8.20	0.25	0.04	-1.68	8.07–8.36	8.10–8.33
	Exp 4	8.60	0.216	2.5	8.60	0.30	0.00	-1.20	8.40–8.80	8.46–8.74
	CHB	13.33	0.512	3.8	13.20	0.60	0.65	-0.05	12.85–13.80	13.00–13.71

IQR = interquartile range; Skew/Kurt. = sample skewness and excess kurtosis; bootstrap CI = percentile interval from 5,000 resamples.

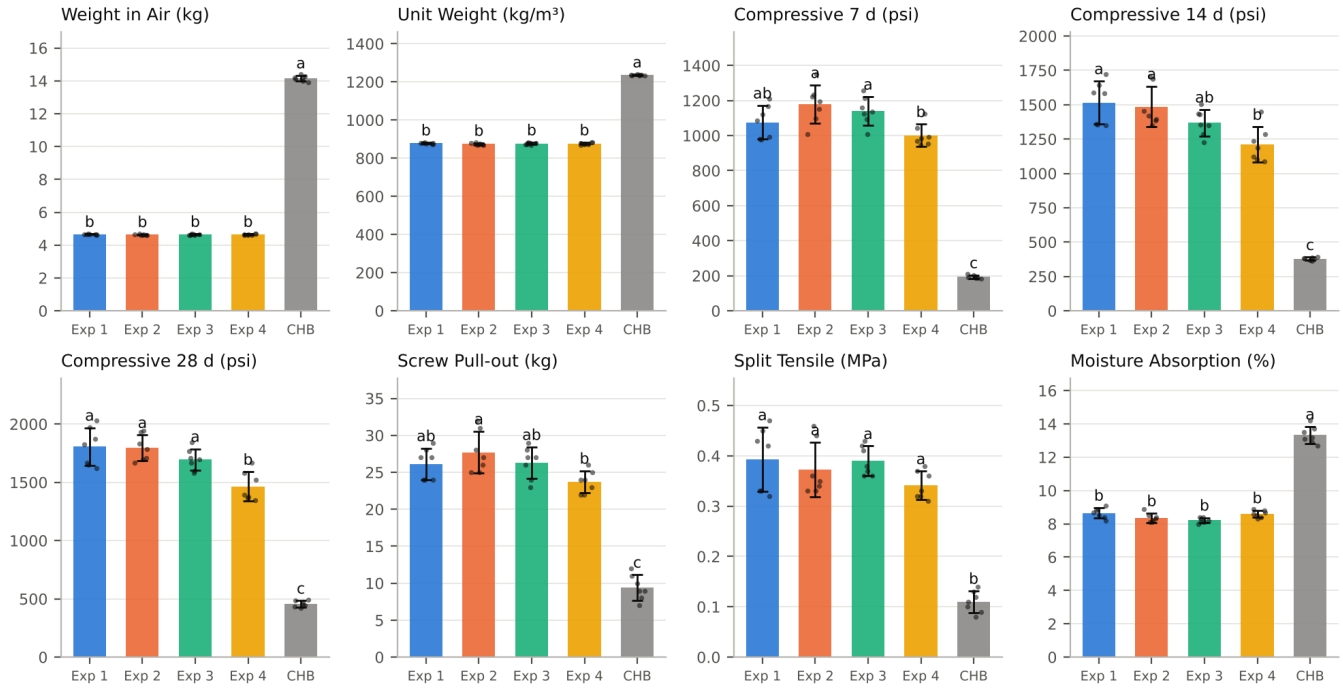


Figure 2. Group means ± 1 SD with individual specimens. Letters: Tukey HSD groupings across all five groups (shared letter = not significantly different).

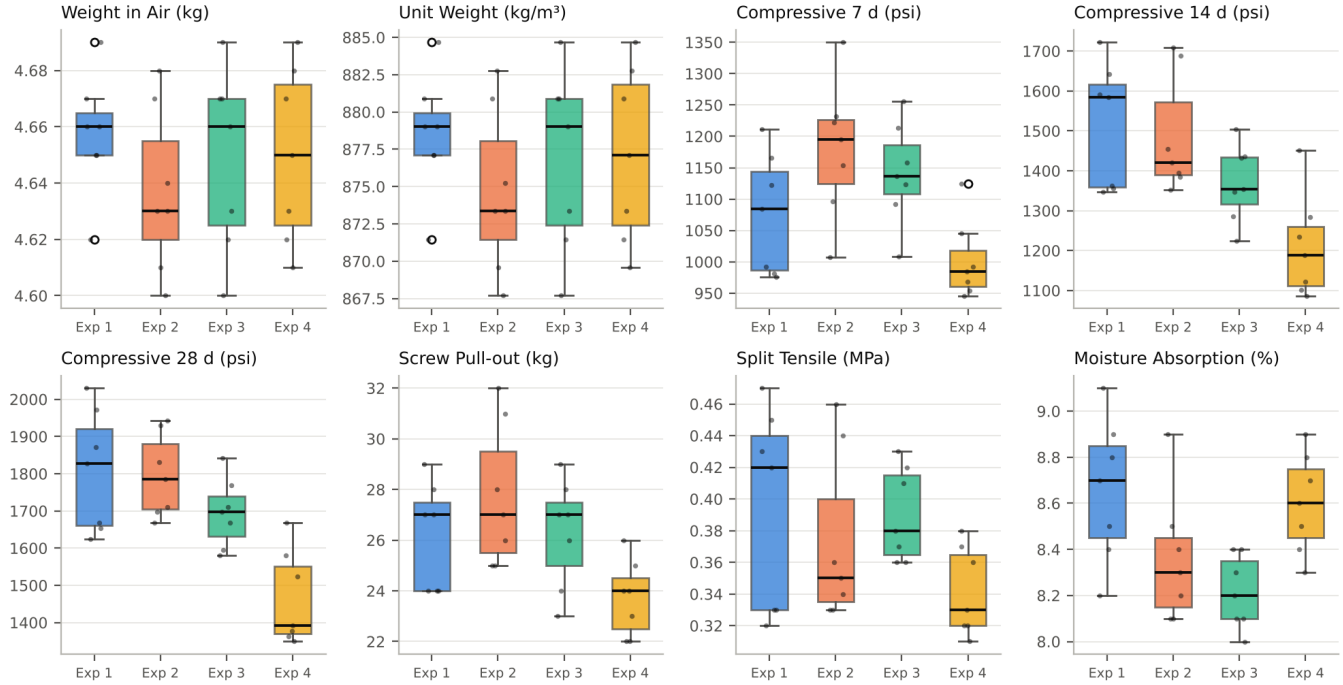


Figure 3. Box plots of the experimental mixes (Exp 1–4): box = IQR, line = median, whiskers = 1.5 × IQR, dots = specimens.

4. Hypothesis Testing

A. One-way ANOVA — all five groups

Test	F	df	p-value	η^2	ω^2	Welch F	Welch p	K–W H	K–W p
Weight in Air (kg)	19266.18	4, 30	< 0.0001	1.000	1.000	4684.1	< 0.0001	17.31	0.0017
Unit Weight (kg/m³)	7144.91	4, 30	< 0.0001	0.999	0.999	13522.7	< 0.0001	17.31	0.0017
Compressive 7 d (psi)	186.83	4, 30	< 0.0001	0.961	0.955	671.0	< 0.0001	23.87	< 0.0001
Compressive 14 d (psi)	106.73	4, 30	< 0.0001	0.934	0.924	375.3	< 0.0001	24.10	< 0.0001
Compressive 28 d (psi)	177.44	4, 30	< 0.0001	0.959	0.953	573.6	< 0.0001	25.67	< 0.0001
Screw Pull-out (kg)	89.70	4, 30	< 0.0001	0.923	0.910	102.1	< 0.0001	22.24	0.0002
Split Tensile (MPa)	55.16	4, 30	< 0.0001	0.880	0.861	130.6	< 0.0001	19.68	0.0006
Moisture Absorption (%)	328.24	4, 30	< 0.0001	0.978	0.974	141.5	< 0.0001	23.37	0.0001

B. One-way ANOVA — experimental mixes only (Exp 1–4)

Test	F	df	p-value	η^2	ω^2	Welch p	K–W p	Methods agree?	Decision
Weight in Air (kg)	0.58	3, 24	0.6360	0.067	0.000	0.5974	0.7185	Yes (3/3)	Not significant
Unit Weight (kg/m³)	0.58	3, 24	0.6359	0.067	0.000	0.5973	0.7185	Yes (3/3)	Not significant
Compressive 7 d (psi)	5.38	3, 24	0.0056	0.402	0.319	0.0076	0.0085	Yes (3/3)	Significant
Compressive 14 d (psi)	7.45	3, 24	0.0011	0.482	0.409	0.0068	0.0072	Yes (3/3)	Significant
Compressive 28 d (psi)	11.16	3, 24	< 0.0001	0.582	0.521	0.0012	0.0023	Yes (3/3)	Significant
Screw Pull-out (kg)	4.02	3, 24	0.0189	0.334	0.244	0.0192	0.0294	Yes (3/3)	Significant
Split Tensile (MPa)	1.80	3, 24	0.1735	0.184	0.079	0.0500	0.1658	1/3 significant	Not significant
Moisture Absorption (%)	4.89	3, 24	0.0086	0.379	0.294	0.0075	0.0127	Yes (3/3)	Significant

Shaded = significant differences among Exp 1–4. Welch p for split tensile is 0.0500 (borderline) while ANOVA and Kruskal–Wallis are clearly non-significant. Negative ω^2 reported as 0.

C. ANOVA source tables — Exp 1–4

Test	Source	SS	df	MS	F	p-value
Weight in Air (kg)	Between mixes	0.0014	3	0.0005	0.58	0.6360
	Within mixes	0.0200	24	0.0008		
	Total	0.0214	27			
Unit Weight (kg/m³)	Between mixes	51.2	3	17.1	0.58	0.6359
	Within mixes	710.6	24	29.6		
	Total	761.8	27			
Compressive 7 d (psi)	Between mixes	127,072.7	3	42,357.6	5.38	0.0056
	Within mixes	189,047.4	24	7,877.0		
	Total	316,120.1	27			
Compressive 14 d (psi)	Between mixes	403,386.4	3	134,462.1	7.45	0.0011
	Within mixes	433,206.3	24	18,050.3		
	Total	836,592.7	27			
Compressive 28 d (psi)	Between mixes	527,252.9	3	175,751.0	11.16	< 0.0001
	Within mixes	377,928.9	24	15,747.0		
	Total	905,181.7	27			
Screw Pull-out (kg)	Between mixes	57.8	3	19.3	4.02	0.0189
	Within mixes	115.1	24	4.7976		
	Total	173.0	27			
Split Tensile (MPa)	Between mixes	0.0117	3	0.0039	1.80	0.1735
	Within mixes	0.0520	24	0.0022		
	Total	0.0637	27			
Moisture Absorption (%)	Between mixes	0.9057	3	0.3019	4.89	0.0086
	Within mixes	1.4829	24	0.0618		
	Total	2.3886	27			

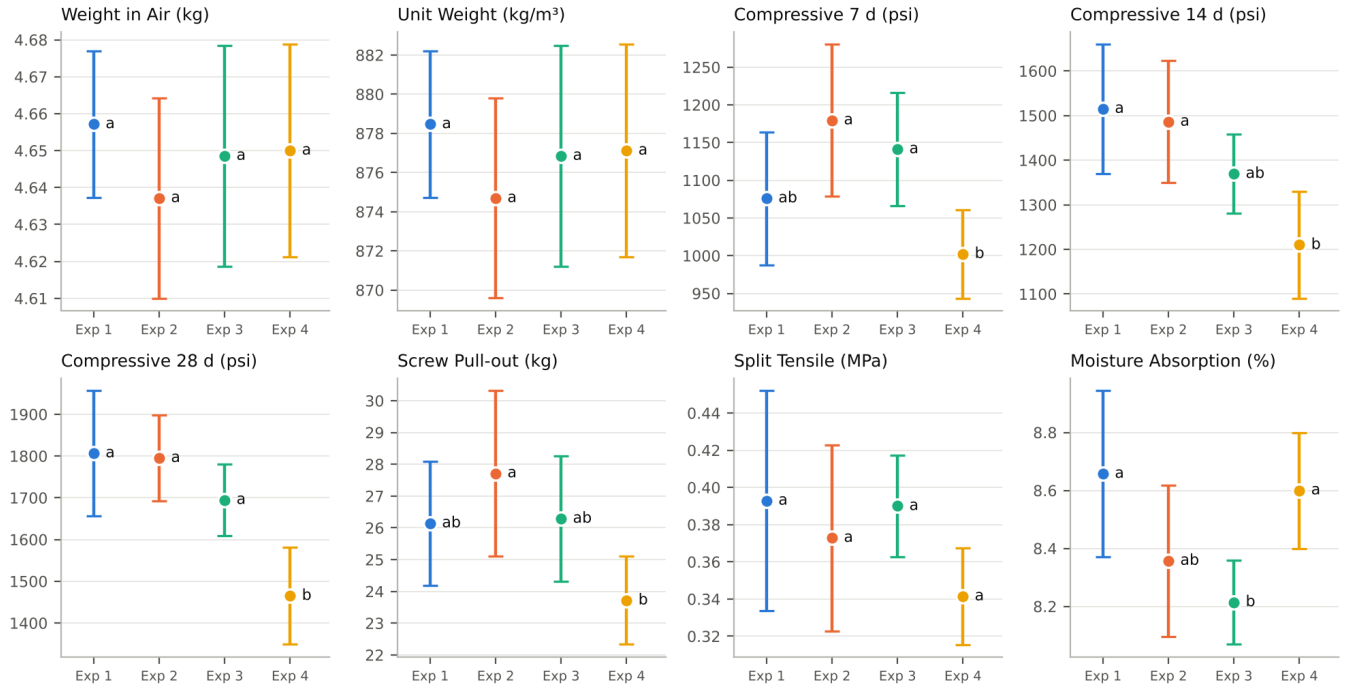


Figure 4. Experimental mixes: means with 95% confidence intervals; letters are Tukey HSD groupings among Exp 1–4.

5. Post-hoc Comparisons

A. Compact letter groupings

Test	Exp 1	Exp 2	Exp 3	Exp 4	CHB	Exp-only letters
Weight in Air (kg)	4.657 b	4.637 b	4.649 b	4.650 b	14.157 a	1:a 2:a 3:a 4:a
Unit Weight (kg/m³)	878.5 b	874.7 b	876.9 b	877.1 b	1236.0 a	1:a 2:a 3:a 4:a
Compressive 7 d (psi)	1076 ab	1179 a	1141 a	1002 b	194 c	1:ab 2:a 3:a 4:b
Compressive 14 d (psi)	1515 a	1486 a	1369 ab	1210 b	380 c	1:a 2:a 3:ab 4:b
Compressive 28 d (psi)	1806 a	1795 a	1695 a	1465 b	457 c	1:a 2:a 3:a 4:b
Screw Pull-out (kg)	26.1 ab	27.7 a	26.3 ab	23.7 b	9.4 c	1:ab 2:a 3:ab 4:b
Split Tensile (MPa)	0.393 a	0.373 a	0.390 a	0.341 a	0.110 b	1:a 2:a 3:a 4:a
Moisture Absorption (%)	8.66 b	8.36 b	8.21 b	8.60 b	13.33 a	1:a 2:ab 3:b 4:a

Mean followed by all-group Tukey letter; last column gives letters from the Exp 1–4 analysis (1 = Exp 1 … 4 = Exp 4).

B. Pairwise comparisons among Exp 1–4: Tukey HSD and Games–Howell

Test	Pair	Mean diff.	Tukey 95% CI	Tukey p	Cohen's d	Games–Howell p	Verdict
Weight in Air (kg)	Exp 1 – Exp 2	0.020	-0.023 to 0.063	0.5736	0.69	0.4915	n.s.
	Exp 1 – Exp 3	0.009	-0.034 to 0.051	0.9440	0.30	0.9346	n.s.
	Exp 1 – Exp 4	0.007	-0.035 to 0.050	0.9663	0.25	0.9571	n.s.
	Exp 2 – Exp 3	-0.011	-0.054 to 0.031	0.8795	-0.40	0.8979	n.s.
	Exp 2 – Exp 4	-0.013	-0.055 to 0.030	0.8379	-0.45	0.8547	n.s.
	Exp 3 – Exp 4	-0.001	-0.044 to 0.041	0.9997	-0.05	0.9998	n.s.
Unit Weight (kg/m³)	Exp 1 – Exp 2	3.8	-4.3 to 11.8	0.5735	0.69	0.4914	n.s.
	Exp 1 – Exp 3	1.6	-6.4 to 9.6	0.9440	0.30	0.9345	n.s.
	Exp 1 – Exp 4	1.3	-6.7 to 9.4	0.9663	0.25	0.9572	n.s.
	Exp 2 – Exp 3	-2.2	-10.2 to 5.9	0.8795	-0.40	0.8978	n.s.
	Exp 2 – Exp 4	-2.4	-10.4 to 5.6	0.8378	-0.45	0.8546	n.s.
	Exp 3 – Exp 4	-0.3	-8.3 to 7.8	0.9997	-0.05	0.9998	n.s.
Compressive 7 d (psi)	Exp 1 – Exp 2	-103	-234 to 27.4	0.1575	-1.17	0.2834	n.s.
	Exp 1 – Exp 3	-65.1	-196 to 65.7	0.5275	-0.73	0.5355	n.s.
	Exp 1 – Exp 4	73.9	-57.0 to 205	0.4211	0.83	0.3672	n.s.
	Exp 2 – Exp 3	38.3	-92.6 to 169	0.8505	0.43	0.8763	n.s.
	Exp 2 – Exp 4	177	46.4 to 308	0.0053	2.00	0.0184	Significant (both)
	Exp 3 – Exp 4	139	8.1 to 270	0.0345	1.57	0.0184	Significant (both)
Compressive 14 d (psi)	Exp 1 – Exp 2	28.7	-169 to 227	0.9779	0.21	0.9842	n.s.
	Exp 1 – Exp 3	146	-52.2 to 344	0.2048	1.09	0.2166	n.s.
	Exp 1 – Exp 4	305	107 to 503	0.0015	2.27	0.0090	Significant (both)
	Exp 2 – Exp 3	117	-81.0 to 315	0.3809	0.87	0.3458	n.s.
	Exp 2 – Exp 4	276	78.2 to 474	0.0040	2.06	0.0138	Significant (both)
	Exp 3 – Exp 4	159	-39.0 to 357	0.1475	1.18	0.0942	n.s.
Compressive 28 d (psi)	Exp 1 – Exp 2	11.3	-174 to 196	0.9983	0.09	0.9987	n.s.
	Exp 1 – Exp 3	112	-73.3 to 297	0.3631	0.89	0.4310	n.s.
	Exp 1 – Exp 4	342	157 to 527	0.0002	2.72	0.0047	Significant (both)
	Exp 2 – Exp 3	100	-84.6 to 285	0.4546	0.80	0.3044	n.s.
	Exp 2 – Exp 4	330	145 to 515	0.0003	2.63	0.0011	Significant (both)
	Exp 3 – Exp 4	230	44.8 to 415	0.0111	1.83	0.0113	Significant (both)
Screw Pull-out (kg)	Exp 1 – Exp 2	-1.6	-4.8 to 1.7	0.5462	-0.72	0.6498	n.s.
	Exp 1 – Exp 3	-0.1	-3.4 to 3.1	0.9993	-0.07	0.9992	n.s.
	Exp 1 – Exp 4	2.4	-0.8 to 5.7	0.1901	1.11	0.1196	n.s.
	Exp 2 – Exp 3	1.4	-1.8 to 4.7	0.6205	0.65	0.7137	n.s.
	Exp 2 – Exp 4	4.0	0.8 to 7.2	0.0113	1.83	0.0363	Significant (both)
	Exp 3 – Exp 4	2.6	-0.7 to 5.8	0.1529	1.17	0.0980	n.s.
Split Tensile (MPa)	Exp 1 – Exp 2	0.020	-0.049 to 0.089	0.8519	0.43	0.9197	n.s.
	Exp 1 – Exp 3	0.003	-0.066 to 0.071	0.9994	0.06	0.9995	n.s.
	Exp 1 – Exp 4	0.051	-0.017 to 0.120	0.1923	1.11	0.2801	n.s.
	Exp 2 – Exp 3	-0.017	-0.086 to 0.051	0.9002	-0.37	0.8800	n.s.
	Exp 2 – Exp 4	0.031	-0.037 to 0.100	0.5940	0.68	0.5483	n.s.
	Exp 3 – Exp 4	0.049	-0.020 to 0.117	0.2336	1.04	0.0356	Games–Howell only
Moisture Absorption (%)	Exp 1 – Exp 2	0.30	-0.07 to 0.67	0.1365	1.21	0.2814	n.s.
	Exp 1 – Exp 3	0.44	0.08 to 0.81	0.0138	1.78	0.0349	Significant (both)
	Exp 1 – Exp 4	0.06	-0.31 to 0.42	0.9727	0.23	0.9772	n.s.
	Exp 2 – Exp 3	0.14	-0.22 to 0.51	0.7076	0.57	0.6582	n.s.
	Exp 2 – Exp 4	-0.24	-0.61 to 0.12	0.2853	-0.98	0.3189	n.s.
	Exp 3 – Exp 4	-0.39	-0.75 to -0.02	0.0366	-1.55	0.0130	Significant (both)

Blue = significant by both procedures; orange = significant by only one. Games–Howell does not assume equal variances. Cohen's d uses √MSW.

C. Each Exp mix versus the CHB control (Dunnett's test)

Test	Exp 1 – CHB	Exp 2 – CHB	Exp 3 – CHB	Exp 4 – CHB
Weight in Air (kg)	-9.500 (-67%) CI -9.612 to -9.388; p < 0.0001	-9.520 (-67%) CI -9.632 to -9.408; p < 0.0001	-9.509 (-67%) CI -9.620 to -9.397; p < 0.0001	-9.507 (-67%) CI -9.619 to -9.395; p < 0.0001
Unit Weight (kg/m³)	-357.5 (-29%) CI -364.5 to -350.6; p < 0.0001	-361.3 (-29%) CI -368.2 to -354.4; p < 0.0001	-359.1 (-29%) CI -366.1 to -352.2; p < 0.0001	-358.9 (-29%) CI -365.8 to -352.0; p < 0.0001
Compressive 7 d (psi)	882 (+455%) CI 773 to 992; p < 0.0001	986 (+508%) CI 876 to 1095; p < 0.0001	947 (+489%) CI 838 to 1057; p < 0.0001	808 (+417%) CI 699 to 918; p < 0.0001
Compressive 14 d (psi)	1135 (+299%) CI 969 to 1301; p < 0.0001	1106 (+291%) CI 941 to 1272; p < 0.0001	989 (+260%) CI 823 to 1155; p < 0.0001	830 (+218%) CI 664 to 996; p < 0.0001
Compressive 28 d (psi)	1349 (+295%) CI 1194 to 1505; p < 0.0001	1338 (+293%) CI 1182 to 1494; p < 0.0001	1238 (+271%) CI 1082 to 1393; p < 0.0001	1008 (+220%) CI 852 to 1163; p < 0.0001
Screw Pull-out (kg)	16.7 (+177%) CI 13.8 to 19.6; p < 0.0001	18.3 (+194%) CI 15.4 to 21.2; p < 0.0001	16.9 (+179%) CI 14.0 to 19.8; p < 0.0001	14.3 (+152%) CI 11.4 to 17.2; p < 0.0001
Split Tensile (MPa)	0.283 (+257%) CI 0.224 to 0.342; p < 0.0001	0.263 (+239%) CI 0.204 to 0.322; p < 0.0001	0.280 (+255%) CI 0.221 to 0.339; p < 0.0001	0.231 (+210%) CI 0.173 to 0.290; p < 0.0001
Moisture Absorption (%)	-4.67 (-35%) CI -5.11 to -4.23; p < 0.0001	-4.97 (-37%) CI -5.41 to -4.53; p < 0.0001	-5.11 (-38%) CI -5.55 to -4.67; p < 0.0001	-4.73 (-35%) CI -5.17 to -4.29; p < 0.0001

Dunnett's test compares every treatment with one control while holding the family-wise error at 5%. All 32 comparisons are significant.

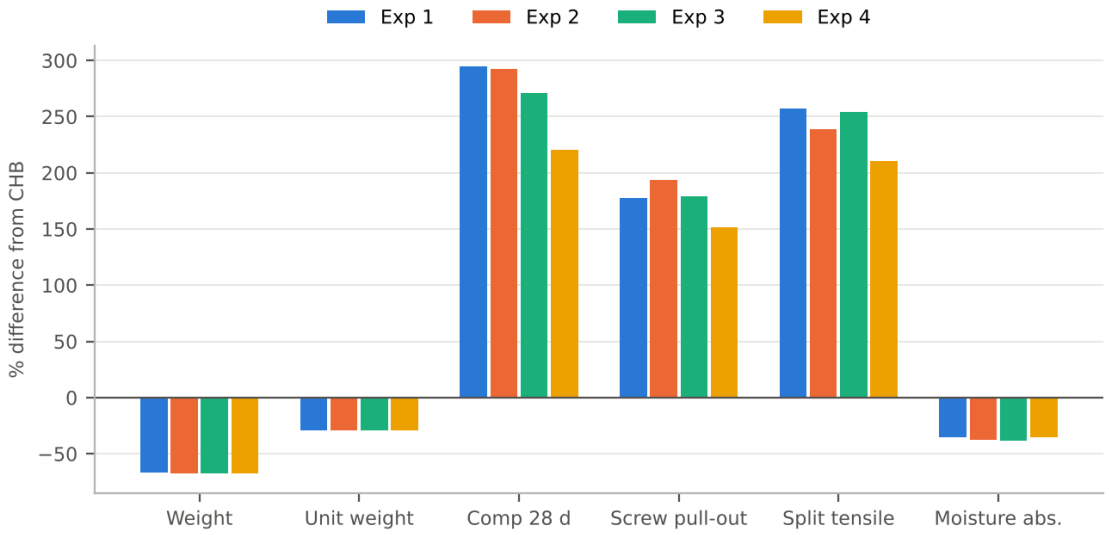


Figure 5. Percentage difference of each Exp mix from CHB. Lower is better for weight, unit weight and moisture absorption; higher is better for strength properties.

6. Effect Size, Statistical Power and Sample Size (Exp 1–4)

Test	η^2	Cohen's f	Effect size	Achieved power	Min. detectable diff. (Tukey)	MDD as % of mean	n per mix for 80% power
Weight in Air (kg)	0.067	0.27	Negligible	17%	0.043	0.9%	—
Unit Weight (kg/m³)	0.067	0.27	Negligible	17%	8.0	0.9%	—
Compressive 7 d (psi)	0.402	0.82	Large	93%	131	11.9%	6
Compressive 14 d (psi)	0.482	0.96	Large	99%	198	14.2%	5
Compressive 28 d (psi)	0.582	1.18	Large	100%	185	10.9%	4
Screw Pull-out (kg)	0.334	0.71	Large	84%	3.2	12.4%	7
Split Tensile (MPa)	0.184	0.47	Large	47%	0.069	18.3%	14
Moisture Absorption (%)	0.379	0.78	Large	91%	0.37	4.3%	6

Power is the probability of detecting the observed effect with n = 7 per mix at $\alpha = 0.05$. Cohen's f: 0.10 small, 0.25 medium, 0.40 large. MDD = smallest difference between two mixes Tukey can detect with this data. For weight and unit weight the true effect is negligible ($\omega^2 < 0$), so a sample-size target is not meaningful.

7. Strength Development Modelling

A. Two-way ANOVA (mix × curing age)

All five groups (n = 105)

Source	SS	df	F	p-value	Partial η^2
Mix	19,455,847	4	435.31	< 0.0001	0.951
Curing age	4,828,886	2	216.08	< 0.0001	0.828
Mix × Age	482,804	8	5.40	< 0.0001	0.324
Residual	1,005,622	90			

Exp 1–4 only (n = 84)

Source	SS	df	F	p-value	Partial η^2
Mix	885,936	3	21.26	< 0.0001	0.470
Curing age	4,883,435	2	175.77	< 0.0001	0.830
Mix × Age	171,775	6	2.06	0.0685	0.147
Residual	1,000,183	72			

B. Strength gain and fitted models

Group	7 d	14 d	28 d	7 d / 28 d	14 d / 28 d	Log model R²	Hyperbolic a, b	Proj. 56 d*	Proj. 90 d*
Exp 1	1076	1515	1806	59.6%	83.9%	0.83	6.07, 0.78	2026	2124
Exp 2	1179	1486	1795	65.7%	82.8%	0.83	5.10, 0.82	1975	2053
Exp 3	1141	1369	1695	67.3%	80.8%	0.87	5.01, 0.82	1861	1933
Exp 4	1002	1210	1465	68.4%	82.6%	0.77	4.66, 0.83	1598	1655
CHB	194	380	457	42.4%	83.1%	0.93	9.67, 0.65	553	600
ACI 209 ref.				70.4%	88.1%		4.00, 0.85		

Strengths in psi. Log model: f(t) = a + b·ln t fitted to individual specimens. Hyperbolic model: f(t) = f28 · t / (a + b·t), the ACI 209 form, fitted to the means (ACI 209 reference for moist-cured Type I cement: a = 4.0, b = 0.85). *Projections beyond 28 days are extrapolations and should be confirmed by testing.

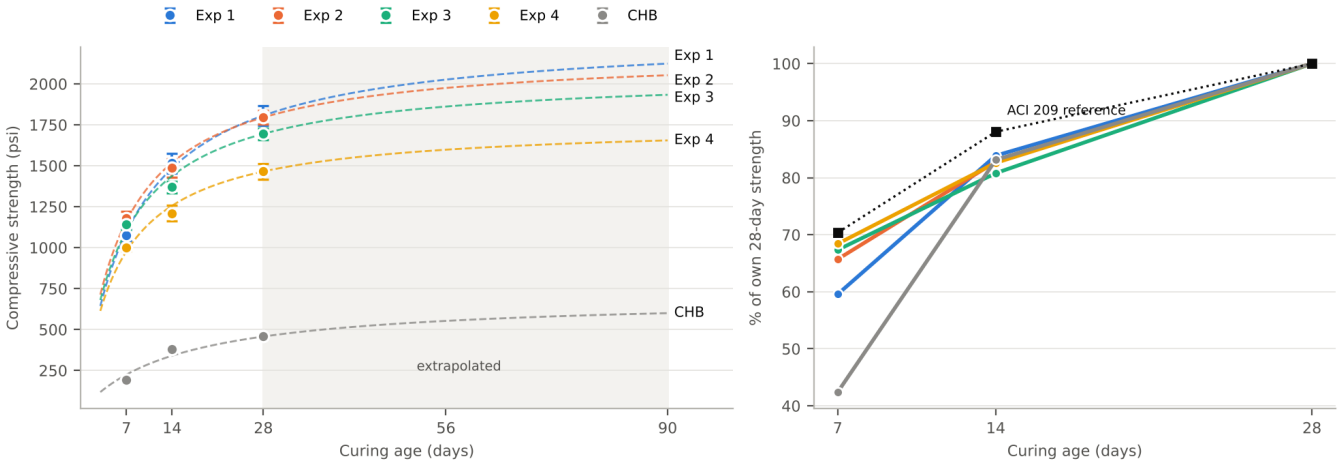


Figure 6. Left: measured means (± SE) with fitted hyperbolic curves (dashed); shaded region is extrapolated. Right: strength as a percentage of each group's own 28-day strength compared with the ACI 209 reference curve.

8. Relationships Between Properties (Exp 1–4 specimens, n = 28)

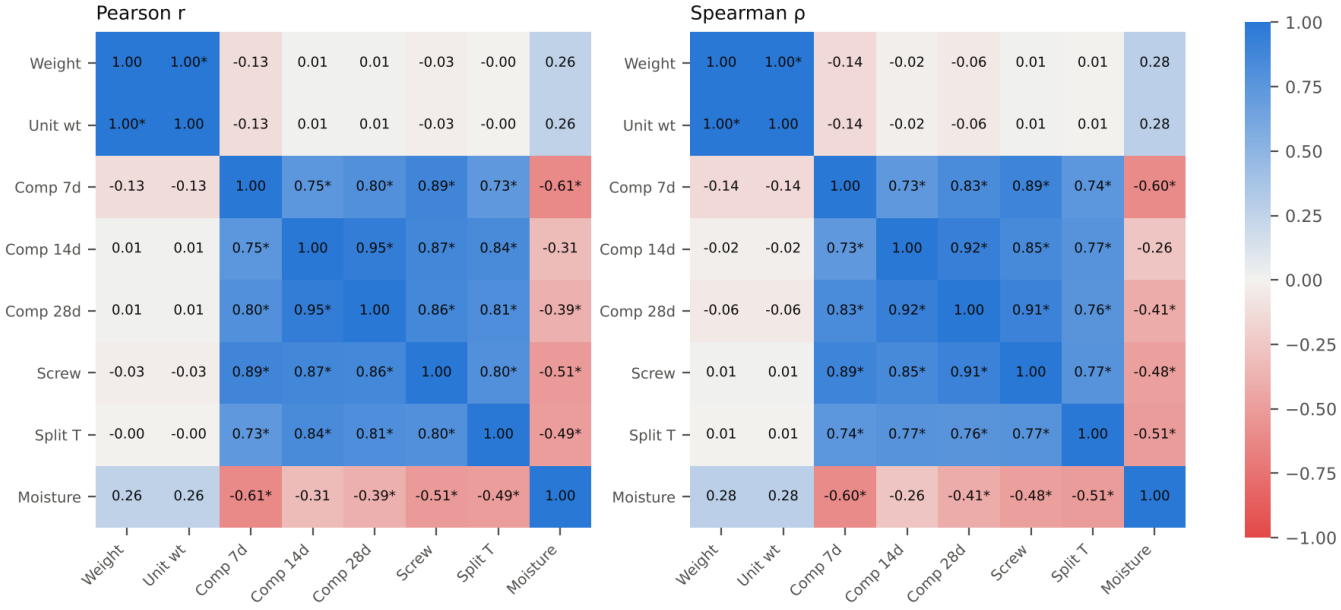


Figure 7. Pearson (linear) and Spearman (rank) correlation matrices. * = significant at α = 0.05 (|r| > 0.374 for n = 28).

Regression models

Model	Equation	R²	p-value	RMSE
28-d strength from 7-d strength	C28 = 209 + 1.347 × C7	0.634	< 0.0001	109
28-d strength from 14-d strength	C28 = 308 + 0.991 × C14	0.908	< 0.0001	54.6
Screw pull-out from 28-d strength	Screw = 5.84 + 0.0119 × C28	0.742	< 0.0001	1.26
Split tensile from 28-d strength	Split = 0.012 + 0.000214 × C28	0.652	< 0.0001	0.0281
28-d strength from moisture absorption	C28 = 3701 -237.7 × Moisture	0.149	0.0424	166
Multiple: C28 from C7 and moisture	C28 = -802 + 1.512 C7 + 98.1 M	0.650 (adj. 0.621)	< 0.0001	

Multiple regression coefficients: C7 p = < 0.0001; moisture p = 0.297 (not significant once 7-day strength is known). Units: C in psi, screw in kg, split in MPa, moisture in %.

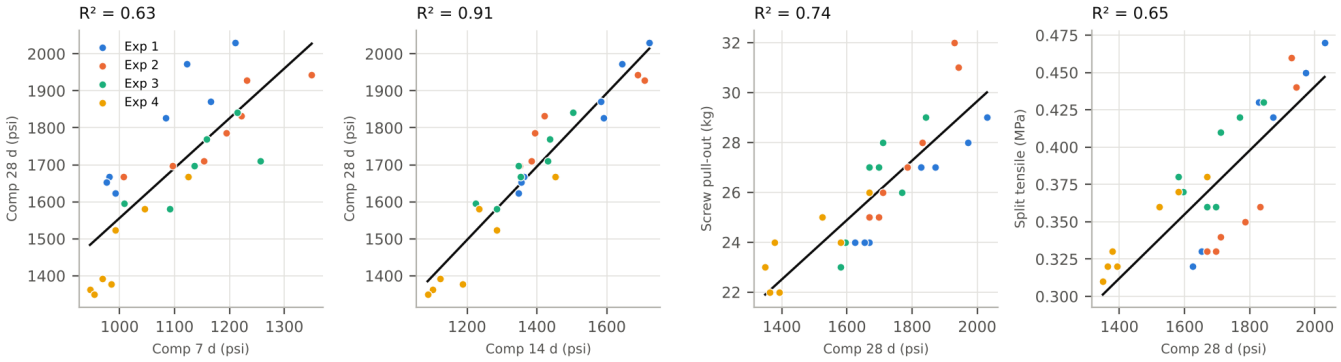


Figure 8. Specimen-level scatter plots with least-squares regression lines.

9. Multivariate Analysis

A. MANOVA (28-d compression, split tensile, screw pull-out, moisture absorption, unit weight)

Statistic	Exp 1–4: value	F	df	p-value	All groups: value	F	p-value
Wilks' lambda	0.0556	6.85	15, 55.6	< 0.0001	9.713e-05	66.31	< 0.0001
Pillai's trace	1.6449	5.34	15, 66.0	< 0.0001	2.33	8.09	< 0.0001
Hotelling-Lawley trace	6.5175	8.33	15, 32.9	< 0.0001	1083	1353.00	< 0.0001
Roy's greatest root	4.9084	21.60	5, 22.0	< 0.0001	1078	6250.63	< 0.0001

B. Principal component analysis

Variable	PC1 loading	PC2 loading
Unit weight	-0.036	0.816
Comp 7 d	0.424	-0.115
Comp 14 d	0.429	0.204
Comp 28 d	0.434	0.164
Screw pull-out	0.442	0.051
Split tensile	0.417	0.082
Moisture abs.	-0.277	0.493
Variance explained	**66.0%**	**16.8%**

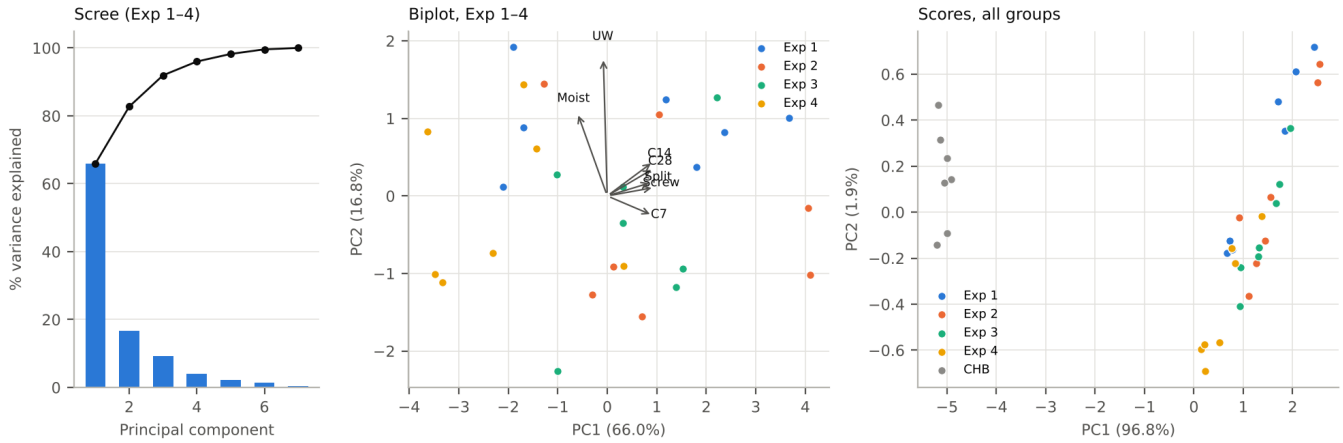


Figure 9. PCA on standardised specimen data. Left: scree plot (bars) and cumulative variance (line). Centre: Exp 1–4 biplot with variable loadings (arrows). Right: all five groups (PC1 explains 96.8% because CHB is far from every Exp mix).

C. Linear discriminant analysis (leave-one-out cross-validation, Exp 1–4)

True \ Predicted	Exp 1	Exp 2	Exp 3	Exp 4	Correct
Exp 1	7	0	0	0	100%
Exp 2	1	5	1	0	71%
Exp 3	0	1	3	3	43%
Exp 4	0	1	1	5	71%

Overall leave-one-out classification accuracy: **71.4%** (chance level 25%). Predictors: compression at 7, 14 and 28 days, screw pull-out, split tensile, moisture absorption.

D. Hierarchical clustering of mix profiles (Ward linkage, standardised means)

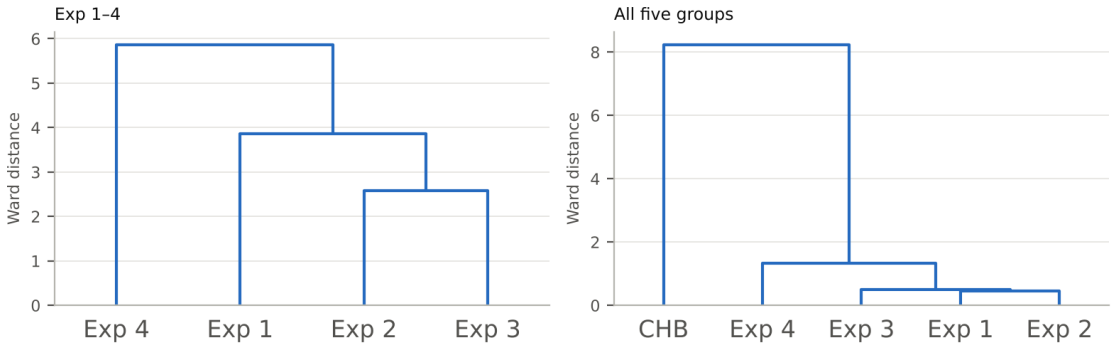


Figure 10. Dendrograms: mixes joined at a lower height have more similar overall property profiles.

10. Benchmarking and Efficiency

A. 28-day compressive strength against reference requirements

Group	Mean (MPa)	Lowest specimen (MPa)	ASTM C129 (4.14 avg / 3.45 ind.)	PNS 16 load-bearing (6.9)*	ASTM C90 (13.8 avg / 12.4 ind.)
Exp 1	12.45	11.20	Meets	Meets	Below
Exp 2	12.38	11.50	Meets	Meets	Below
Exp 3	11.68	10.90	Meets	Meets	Below
Exp 4	10.10	9.30	Meets	Meets	Below
CHB	3.15	2.90	Below	Below	Below

Conversion: 1 psi = 0.006895 MPa. ASTM C129 (non-load-bearing) and C90 (load-bearing) specify net-area strength for the average of 3 units and individual units. *PNS 16 values are the commonly cited Philippine requirements; confirm against the edition used. The workbook does not state whether strengths are on gross or net area, so this is an indicative comparison.

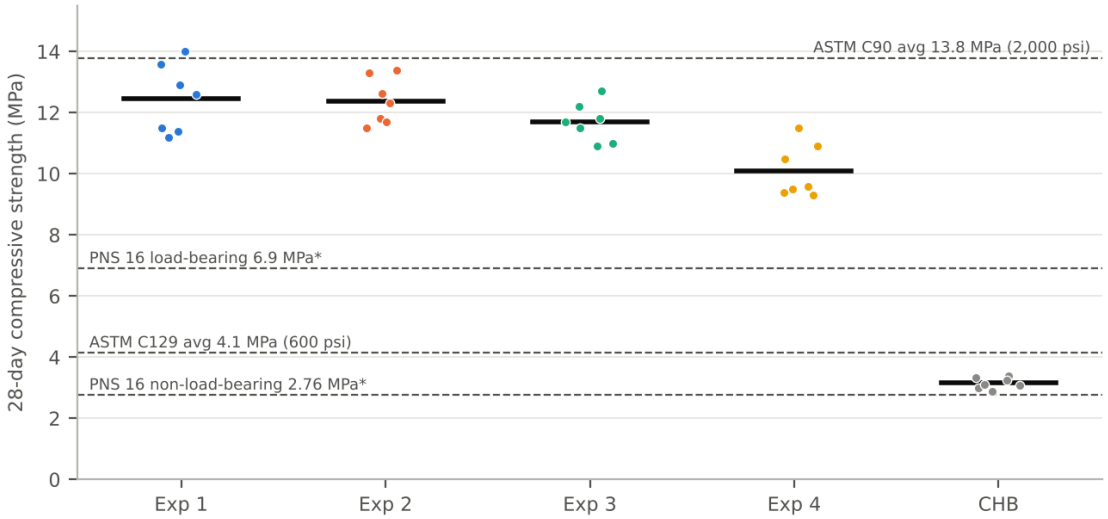


Figure 11. Individual 28-day specimens (dots) and group means (bars) against reference levels (dashed lines).

B. Efficiency ratios

Group	Specific strength (psi per kg/m³)	Split tensile / compressive	Weight reduction vs CHB	28-d strength vs CHB
Exp 1	2.056 ± 0.184	3.14%	67.1%	3.95×
Exp 2	2.053 ± 0.130	3.00%	67.2%	3.93×
Exp 3	1.933 ± 0.101	3.34%	67.2%	3.71×
Exp 4	1.670 ± 0.144	3.38%	67.2%	3.20×
CHB	0.370 ± 0.022	3.52%	—	—

11. Decision Analysis

A. Composite performance index (Exp 1–4)

Group	Comp 28 d	Split tensile	Screw pull-out	Low absorption	Low unit weight	Index	Rank	z-score index
Exp 1	1.00	1.00	0.61	0.00	0.00	0.521	3	+0.02
Exp 2	0.97	0.61	1.00	0.68	1.00	0.851	1	+0.39
Exp 3	0.67	0.94	0.64	1.00	0.43	0.738	2	+0.26
Exp 4	0.00	0.00	0.00	0.13	0.36	0.097	4	-0.67

Each property is scored 0 (worst Exp mix) to 1 (best Exp mix) and averaged with equal weights. The z-score index is an alternative that measures distances in standard-deviation units, so small differences are not exaggerated.

B. Weighting scenarios

Scenario	Weights (C28/Split/Screw/Absorp./UW, %)	Exp 1	Exp 2	Exp 3	Exp 4	Best
Equal weights	20/20/20/20/20	0.52	0.85	0.74	0.10	Exp 2
Strength-focused	50/20/20/5/5	0.82	0.89	0.73	0.02	Exp 2
Durability-focused	20/10/10/50/10	0.36	0.79	0.84	0.10	Exp 3
Lightweight-focused	20/10/10/10/50	0.36	0.92	0.61	0.19	Exp 2
Fastening-focused	20/10/50/10/10	0.60	0.92	0.69	0.05	Exp 2

C. Monte Carlo sensitivity (20,000 random weight sets)

Group	Ranked 1st	Ranked last	Mean rank
Exp 1	2.9%	3.9%	2.87
Exp 2	74.4%	0.0%	1.30
Exp 3	22.7%	0.0%	1.87
Exp 4	0.0%	96.1%	3.96

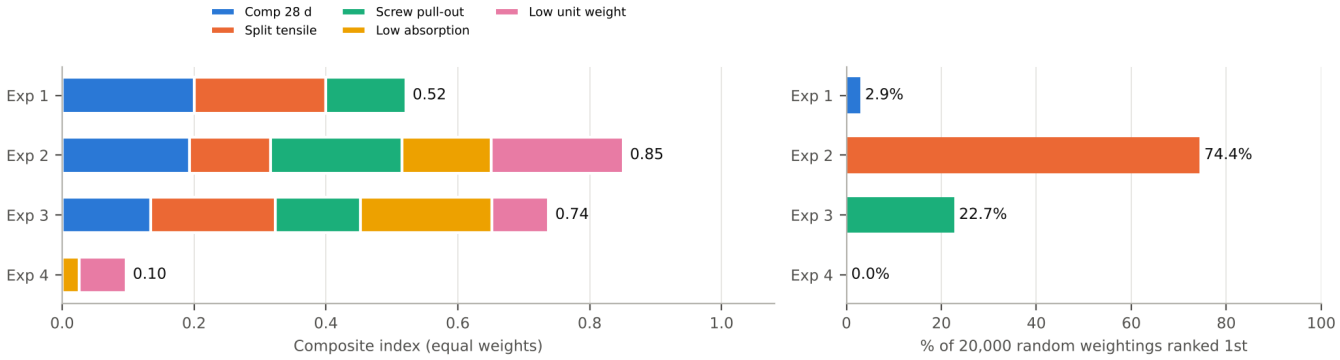


Figure 12. Left: composite index with each property's contribution. Right: share of random weightings in which each mix ranks first.

12. Explanations and Interpretation

12.1 What each method does and how to read it

Mean, SD, CV. The mean is the average of the 7 replicates; the standard deviation (SD) is the typical distance of a specimen from that mean; the coefficient of variation (CV = SD/mean × 100) expresses that scatter as a percentage. CV under ~10% indicates consistent production and testing.

Median, IQR, skewness, kurtosis. The median and interquartile range describe the centre and middle 50% of the data without being affected by extreme values. Skewness measures asymmetry (0 = symmetric); kurtosis measures tail heaviness (0 = normal). With n = 7 these are rough indicators only.

Confidence intervals (t and bootstrap). A 95% CI is the range likely to contain the true mean. The t-interval assumes normality; the bootstrap interval re-samples the data 5,000 times and makes no distributional assumption. Close agreement between the two supports the reliability of the means.

Grubbs' test and IQR rule. Grubbs' test checks whether the most extreme value in a group is a statistical outlier. The IQR rule flags values beyond 1.5 × IQR from the quartiles; it is more sensitive and is used only as a screen.

Shapiro–Wilk, Levene, Bartlett. Shapiro–Wilk tests normality ($p > 0.05$ = consistent with normal). Levene (robust) and Bartlett (sensitive to non-normality) test whether groups have equal variances. These are the assumptions behind ANOVA.

One-way ANOVA. Tests H_0: all group means are equal. F = between-group variance ÷ within-group variance. $p < 0.05$ means at least one group differs. It does not identify which ones.

Welch ANOVA and Kruskal–Wallis. Welch ANOVA does not assume equal variances. Kruskal–Wallis compares ranks and does not assume normality. When these agree with ordinary ANOVA, the conclusion is robust.

Tukey HSD, Games–Howell, Dunnett. Post-hoc tests that compare groups while keeping the overall false-positive rate at 5%. Tukey compares all pairs assuming equal variances; Games–Howell does the same without that assumption; Dunnett compares each treatment only with a control (CHB), which gives it more power for that question.

Effect sizes (η^2, ω^2, Cohen's d, Cohen's f). η^2 is the proportion of variation explained by mix; ω^2 is its less-biased version. Cohen's d expresses a pairwise difference in SD units (0.8+ = large). Cohen's f is used for power analysis (0.40+ = large).

Power and minimum detectable difference. Power is the chance of detecting a real effect of the observed size. 80% is the usual target. The minimum detectable difference is the smallest gap between two mixes that the experiment could reliably detect.

Two-way ANOVA. Tests mix, curing age and their interaction together. A significant interaction means the mixes gain strength at different rates.

Strength development models. The log model describes strength as rising with the logarithm of age. The hyperbolic model (ACI 209 form) describes strength approaching a limit; its b value indicates how close the material is to its final strength at 28 days.

Correlation and regression. Pearson r measures linear association; Spearman ρ measures monotonic (rank) association and is resistant to outliers. Regression gives an equation to predict one property from another; R^2 is the share of variation it explains and RMSE is its typical prediction error.

MANOVA. Tests differences across several properties at once, taking their correlations into account. Wilks' λ close to 0 indicates strong separation of groups.

PCA. Combines correlated properties into a few uncorrelated components. Loadings show which properties drive each component; the biplot shows where each specimen sits.

Discriminant analysis. Asks how accurately the mix can be identified from a specimen's measured properties. Leave-one-out validation tests each specimen with a model built without it.

Clustering. Groups mixes with similar overall property profiles; the height at which they join measures dissimilarity.

Composite index and sensitivity analysis. Combines several properties into one score for ranking. Because the result depends on how the properties are weighted, scenario and Monte Carlo analyses test whether the ranking holds under different priorities.

12.2 Interpretation of the results

Data quality (Section 2). The data set is clean and well suited to the analyses. No value is a statistical outlier (all Grubbs $G \leq 1.94$ vs critical 2.02). The four mild IQR flags are ordinary scatter in samples of seven and were kept. Normality holds in 38 of 40 group samples, and the two marginal cases (Exp 2 at 14 days and Exp 2 split tensile, $p \approx 0.03$) are caused by one or two high values and do not change any conclusion. Variances are equal among Exp 1–4 for every test. Variance differences appear only when CHB is included (CHB compressive strength scatters far less in absolute terms because its values are much smaller), and the Welch and Games–Howell methods handle that case.

Variability (Section 3). Physical properties are very uniform (CV 0.2–1.2% for weight and unit weight), confirming consistent block dimensions and mix proportioning. Mechanical properties show normal scatter for masonry units (CV 5–11% in compression), with split tensile strength and CHB screw pull-out the most variable (CV 15–20%). The t and bootstrap confidence intervals agree closely in every case.

Differences among the mixes (Sections 4–5). When CHB is included, every test is significant with very large effects (η^2 0.88–1.00), which only confirms that CHB is a different material. The informative comparison is among Exp 1–4. Weight and unit weight do not differ ($p = 0.64$), so all four mixes produce blocks of the same mass. Compressive strength differs at every age, with the effect growing over time ($\eta^2 = 0.40 \rightarrow 0.48 \rightarrow 0.58$). At 28 days Exp 4 is lower than each of the other three by 230–342 psi ($d = 1.8$–2.7), while Exp 1, Exp 2 and Exp 3 cannot be separated. Screw pull-out differs only between Exp 2 and Exp 4. Moisture absorption is lowest for Exp 3, significantly lower than Exp 1 and Exp 4. For split tensile strength, ANOVA and Kruskal–Wallis find no difference; Games–Howell flags Exp 3 > Exp 4 ($p = 0.036$) because both groups have small variances. This is weak evidence that Exp 4 is also lower in tension. Tukey, Games–Howell and Kruskal–Wallis otherwise agree on every conclusion.

Comparison with CHB (Section 5C). Dunnett's test confirms that each Exp mix is better than CHB on every property (all 32 $p < 0.0001$). The improvement is very large, not merely significant: the Exp blocks weigh about one-third as much as CHB, yet carry 3.2–4.0 times the compressive load, hold 2.5–2.9 times the screw load, and absorb about one-third less water.

Power (Section 6). For 28-day compression the experiment has almost 100% power and could detect a difference of about 185 psi (11% of the mean). The non-significant split tensile result has only 47% power: the experiment could only detect a difference of about 0.069 MPa (18%). The absence of a significant difference is therefore inconclusive rather than proof that the mixes are equal; about 14 specimens per mix would be needed. For weight the differences are genuinely negligible ($\leq$ 0.02 kg).

Strength development (Section 7). Age and mix both have very large effects (partial $\eta^2 \approx 0.83$ and 0.47 among the Exp mixes). The Exp mixes develop strength in parallel (interaction $p = 0.068$), so their ranking at 7 days already predicts their ranking at 28 days; Exp 4 is lowest throughout. The Exp mixes reach 60–68% of their 28-day strength by 7 days, slightly below the ACI 209 reference (70%), and 81–84% by 14 days. CHB develops strength very differently: only 42% at 7 days, then a large jump by 14 days. This explains the significant interaction when CHB is included. The fitted models ($R^2 =$ 0.77–0.93) project modest further gains to about 1655–2124 psi at 90 days for the Exp mixes. These are projections only and need confirmation by testing. Exp 1 has the strongest late gain (+68% from 7 to 28 days), which is why it overtakes Exp 2 after the first week.

Relationships (Section 8). Among the Exp specimens, all mechanical properties rise together. 14-day strength predicts 28-day strength very well ($R^2 = 0.91$, typical error 55 psi), and 7-day strength predicts it moderately ($R^2 = 0.63$, error 109 psi). This supports using 14-day tests for early quality control. Screw pull-out increases by about 1.2 kg per 100 psi of compressive strength ($R^2 = 0.74$). Split tensile increases by about 0.021 MPa per 100 psi ($R^2 = 0.65$). Higher moisture absorption goes with lower strength ($r = -0.39$ at 28 days), consistent with porosity weakening the block, but once early strength is known, moisture adds no predictive information. Block weight is unrelated to strength ($r \approx 0$): within this narrow weight range, strength is controlled by the mix, not by mass. Spearman and Pearson coefficients agree, so the relationships are not driven by individual extreme specimens.

Multivariate view (Section 9). MANOVA confirms that the four Exp mixes differ when all properties are considered together (Pillai's trace $p < 0.0001$). PCA shows that one dimension, overall mechanical quality (PC1, 66% of variance), dominates. Compression, screw pull-out and split tensile all load on it and moisture absorption loads against it. The second dimension (PC2, 17%) is essentially unit weight, which is independent of strength. Exp 4 sits on the low-quality side of PC1. Discriminant analysis identifies the mix of a specimen correctly 71% of the time (vs 25% by chance). Exp 1 is classified perfectly, while Exp 3 and Exp 4 are most often confused because of overlapping early-age and tensile values. Clustering puts Exp 2 and Exp 3 closest, then Exp 1, with Exp 4 the most distinct Exp mix and CHB far from all of them.

Benchmarking and efficiency (Section 10). All Exp mixes exceed the ASTM C129 non-load-bearing requirement by a wide margin (more than twice the 4.14 MPa average), and their means are above the commonly cited PNS 16 load-bearing level of 6.9 MPa. They remain below the ASTM C90 load-bearing average of 13.8 MPa: Exp 1, the best mix, reaches 12.5 MPa (about 90% of it). CHB averages 3.15 MPa, below ASTM C129. The Exp mixes are 4.5–5.6 times more efficient than CHB in strength per unit weight, which is their main engineering advantage. These comparisons are indicative because the area

basis (gross or net) and the block geometry are not recorded in the workbook.

Decision analysis (Section 11). Exp 2 ranks first under equal weights and under the strength-, lightweight- and fastening-focused scenarios. Exp 3 ranks first only when durability (low absorption) dominates. Across 20,000 random weightings, Exp 2 is best 74% of the time and Exp 3 23% of the time, and Exp 4 is last 96% of the time. The z-score index gives the same order (Exp 2 > Exp 3 > Exp 1 > Exp 4). The recommendation is therefore robust to how the criteria are weighted. Exp 1 ranks below Exp 3 mainly because of its higher water absorption and slightly heavier blocks, even though it is the strongest.

12.3 Conclusions

- The experimental blocks are statistically and practically superior to the conventional CHB in every property tested, combining about two-thirds lower weight with three to four times higher strength and lower water absorption.
- Exp 4 is the least effective experimental mix: it has the lowest compressive strength at every age, the lowest screw pull-out and the lowest specific strength, and it ranks last under almost every weighting.
- Exp 1, Exp 2 and Exp 3 are statistically equivalent in 28-day compressive strength; their differences lie in secondary properties.
- **Exp 2 is the recommended mix**: highest composite score, highest screw pull-out, lowest unit weight, strong early strength, and first place in about three-quarters of all weighting scenarios.
- Exp 3 is the preferred alternative where durability and moisture resistance govern; Exp 1 where maximum 28-day compressive strength is the sole criterion.
- 14-day compressive strength is a reliable early predictor of 28-day strength ($R^2 \approx 0.91$) and can be used for faster quality control.

12.4 Recommendations for further testing

- Increase split tensile replicates to about 14 per mix to settle whether the mixes differ in tensile strength.
- Test compressive strength at 56 and 90 days to verify the projected later-age gains, particularly for Exp 1.
- Record the loaded area (gross or net) and the block dimensions so results can be compared formally with ASTM C90/C129 and PNS 16.
- If load-bearing use is intended, investigate adjustments to Exp 1 or Exp 2 to reach the ASTM C90 average of 13.8 MPa (about a 10% increase needed).

12.5 Limitations

- Seven specimens per group limit the power to detect small differences, especially in split tensile strength.
- Specimens at 7, 14 and 28 days are treated as independent (compression testing is destructive); specimen-level correlations assume replicate i refers to the same specimen across sheets, following the Raw Files layout.
- The Moisture Absorption sheet has no matching table in Raw Files; its values were used as given.
- Unit weight is directly proportional to weight in air for the Exp blocks (r = 1.00), so the two measures are not independent evidence.
- Later-age strength projections and standards comparisons are indicative and require confirmation.